SEMESTER-VII

S.No.	Course Code	Name of the Course		L	T	P	C
1	V23CST15	Deep Learning	PCC	2	1	0	3
2	V23MBT57	Human Resources & Project Management	PCC	2	0	0	2
3	**Professional Elective-IV**		PEC	3	0	0	3
	V23CSTPE13	Software Architecture & Design Patterns					
4	**Professional Elective-V**		PEC	3	0	0	3
	V23CSTPE17	Generative AI					
5	**Open Elective -III**		OEC	0	0	6	3
	V23AITOE08	Java Full Stack Technologies					
6	**Open Elective -IV**		OEC	0	0	6	3
	V23AITOE09	Agentic AI					
7	V23CSSE05	Prompt Engineering / SWAYAM Plus - Certificate program in Prompt Engineering and ChatGPT	SEC	0	1	2	2
8	V23CST16	AI Ethics and Cyber Laws	Audit Course	2	0	0	0
		Evaluation of Industry Internship / Mini Project	Internship	-	-	-	2
Total				18	02	02	21

Semester	VII	L	T	P	C	COURSE CODE
Regulation	V23	2	1	0	3	V23CST15
Name of the Course	Deep Learning					
Branch	Common to CSE & CST					

Course Outcomes: After successful completion of the course, the student will be able to:

CO No.	Course Outcome	Knowledge Level
CO1	Explain the fundamentals of Machine Learning and Deep Learning, including data preprocessing and evaluation techniques.	[K2]
CO2	Illustrate the working and training process of Artificial Neural Networks and Deep Neural Networks with optimization techniques	[K2]
CO3	Develop ANN models for classification tasks using Keras and model building pipeline	[K3]
CO4	Develop CNNs and RNNs using PyTorch for solving real-world problems	[K3]
CO5	Illustrate applications of Deep Learning in computer vision, natural language processing, and sequence models	[K2]

SYLLABUS

UNIT I: Fundamentals of Deep Learning: Artificial Intelligence, History of Machine learning: Probabilistic Modeling, Early Neural Networks, Kernel Methods, Decision Trees, Random forests and Gradient Boosting Machines, Fundamentals of Machine Learning: Four Branches of Machine Learning, Evaluating Machine learning Models, Overfitting and Underfitting. **[Text Book 2]**

UNIT II: Introducing Deep Learning: Biological and Machine Vision, Human and Machine Language, Artificial Neural Networks, Training Deep Networks, Improving Deep Networks. **[Text Book3]**

UNIT III: Neural Networks: Anatomy of Neural Network, Introduction to Keras: Keras, TensorFlow, Theano and CNTK, Setting up Deep Learning Workstation, Classifying Movie Reviews: Binary Classification, Classifying newswires: Multiclass Classification. **[Text Book 2]**

UNIT IV: Convolutional Neural Networks: Nerual Network and Representation Learing, Convolutional Layers, Multichannel Convolution Operation, Recurrent Neural Networks: Introduction to RNN, RNN Code, PyTorch Tensors: Deep Learning with PyTorch, CNN in PyTorch. **[Text Book 3]**

UNIT V: Interactive Applications of Deep Learning: Machine Vision, Natural Language processing, Generative Adversial Networks, Deep Reinforcement Learning. [Text Book 1] Deep Learning Research: Autoencoders, Deep Generative Models: Boltzmann Machines Restricted Boltzmann Machines, Deep Belief Networks. **[Text Book 1]**

Text Books:

1. Deep Learning- Ian Goodfellow, YoshuaBengio and Aaron Courvile, MIT Press, 2016

2. Deep Learning with Python - Francois Chollet, Released December 2017, Publisher(s): Manning Publications, ISBN: 9781617294433

3. Deep Learning Illustrated: A Visual, Interactive Guide to Artificial Intelligence - Jon Krohn, Grant Beyleveld, AglaéBassens, Released September 2019, Publisher(s): Addison-Wesley Professional, ISBN: 9780135116821

4. Deep Learning from Scratch - Seth Weidman, Released September 2019, Publisher(s): O'Reilly Media, Inc., ISBN: 9781492041412

Reference Books:

1. Artificial Neural Networks, Yegnanarayana, B., PHI Learning Pvt. Ltd, 2009.

2. Matrix Computations, Golub, G.,H., and Van Loan,C.,F, JHU Press,2013.

3. Neural Networks: A Classroom Approach, Satish Kumar, Tata McGraw-Hill Education, 2004.

WEB RESOURCES

1. Swayam NPTEL: Deep Learning: https://onlinecourses.nptel.ac.in/noc22_cs22/preview

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Semester	VII	L	T	P	C	COURSE CODE
Regulation	V23	2	0	0	2	V23MBT57
Name of the Course	Human Resources & Project Management					
Branch	Common to CSE & CST					

Course Outcomes: After successful completion of the course, the student will be able to:

CO No.	Course Outcome	Knowledge Level
CO1	**Explain** the concepts, functions, trends and ethical aspects of Human Resource Management, including HR planning, recruitment and selection	[K2]
CO2	**Explain** HRD, HR accounting, training, performance appraisal, career development, counselling and group interaction.	[K2]
CO3	**Explain** the basic concepts, types, life cycle, networks, resource management and monitoring of projects	[K2]
CO4	**Apply** project-selection and appraisal techniques such as SWOT analysis, cash-flow analysis, payback period and net present value	[K3]
CO5	**Explain** project organization, planning, control, implementation, review, performance evaluation and abandonment analysis	[K2]

SYLLABUS

UNIT I: HRM: Nature, Scope, Concept of HRM - Functions of HRM - Role of HR manager – Emerging trends in HRM – E-HRM - HR audit models – ethical aspects of HRM. HR Planning – Demand and Supply forecasting of HR - Job Design- Recruitment - Sources of recruitment -Selection- Selection Procedure.

UNIT –II:HRD – HR accounting – Models - Concept of Training and Development –- Methods of Training. Performance Appraisal: Importance – Methods of performance appraisal – Career Development and Counseling – group interaction.

UNIT –III Basics of Project Management –Concept–resource management- Project environment – Types of Projects –project networks-DPR- Project life cycle – Project proposals – Monitoring project progress – Project appraisal and Project selection-80-20 rules-Production technology-communication matrix

UNIT IV Project Selection and Appraisal: Brainstorming and concept evolution, Project selection and evaluation, Selection criteria and models, Types of appraisals, SWOT analysis, Cash flow analysis, Payback period, and Net present value.

UNIT V: Identify various project types and their unique management challenges and apply appropriate management strategies for each. Project Implementation and Review: Forms of project organization – project planning – project control – human aspects of project management – prerequisites for successful project implementation – project review – performance evaluation – abandonment analysis

Text Books :

1. Robert L. Mathis, John H. Jackson, Manas Ranjan Tripathy, Human Resource Management, Cengage Learning 2016.
2. Sharon Pande and Swapnalekha Basak, Human Resource Management, Text and Cases, Vikas Publishing, 2e, 2016.
3. Stewart R. Clegg, Torgeir Skyttermoen, Anne Live Vaagaasar, Project Management, Sage Publications, 1e, 2021.
4. K. Nagarajan, Project Management, New Age International Publishers, 8e, 2017.

References :

1. Subba Rao P: ―Personnel and Human Resource Management-Text and Cases‖, Himalaya Publications, Mumbai, 2013.
2. K Aswathappa: ―Human Resource and Personnel Management‖, Tata McGraw Hill, New Delhi, 2013.
3. Prasanna Chandra, ―Projects, Planning, Analysis, Selection, Financing Implementation and Review‖, Tata McGraw Hill Company Pvt. Ltd., New Delhi 1998
4. Vasanth Desai, Project Management, 4th edition, Himalaya Publications 2018. 5. Lalithabalakrishnan and Gowri, Project Management, Himalaya publishing house, New Delhi 2022.

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Semester	VII	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE13
Name of the Course	Software Architecture & Design Patterns (Professional Elective-IV)					
Branch	Common to CSE & CST					

Course Outcomes: After successful completion of the course, the student will be able to:

CO No.	Course Outcome	Knowledge Level
CO1	Explain the concept of **design** patterns and their organization for practical use	[K2]
CO2	Apply object oriented analysis and design concepts to solve complex problems	[K3]
CO3	Illustrate the intent and structure of common structural design patterns	[K2]
CO4	**Apply s**oftware design using **MVC architectural pattern**	[K3]
CO5	**Outline** object oriented system on the web	[K2]

SYLLABUS

UNIT I: Introduction: Design pattern, Describing design patterns, the catalog of design pattern, organizing the catalog, how design patterns solve design problems, how to select a design pattern, how to use a design pattern What is object oriented development? key concepts of object oriented design other related concepts, benefits and drawbacks of the paradigm **[Textbook 1& 2]**

UNIT II: Analyzing a System: Overview of the analysis phase, stage 1 gathering the requirements functional requirements specification, defining conceptual classes and relationships, using the knowledge of the domain Design and Implementation, discussions and further reading. **[Textbook 2]**

UNIT III: Design Pattern Catalog: Structural patterns, Adapter, bridge, composite, decorator, facade, flyweight, proxy. **[Textbook 1]**

UNIT IV: Interactive systems and the MVC architecture: Introduction, MVC architectural pattern, Analyzing a simple drawing program, Designing the system, Designing of the subsystems, Getting into implementation, Implementing undo operation, Drawing incomplete items, Adding a new feature, Pattern based solutions. **[Textbook 2]**

UNIT V: Designing with Distributed Objects: Client-server systems, Java remote method invocation, Implementing an object oriented system on the web, Web services (SOAP, Restful), Enterprise Service Bus. **[Textbook 2]**

Text Books:

1. Design patterns - Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides, 2013, Pearson Publisher.
2. Object oriented analysis, design and implementation - Brahma Dathan, Sarnath Rammath , 2015 Edition, universities press.

Reference Books:

1. Pattern Oriented Software Architecture - Frank Buschmann, Regine Meunier , Hans Rohnert. Peter Sommerlad, 1996, Wiley Publisher.
2. Anti Patterns: Refactoring Software, Architectures and Projects in Crisis- William J Brown 1998, JohnWiley and sons Publisher.,

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Semester	VII	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE17
Name of the Course	Generative AI(Professional Elective-V)					
Branch	Common to CSE & CST					

Course Outcomes: After successful completion of the course, the student will be able to:

CO No.	Course Outcome	Knowledge Level
CO1	Apply fundamental concepts of Generative AI to experiment with generative models and analyze prompt and parameter effects.	[K3]
CO2	Apply transformer-based language models and prompt engineering techniques to design text generation systems	[K3]
CO3	Analyze image generation models based on quality, stability, and prompt alignment	[K4]
CO4	Examine Generative models for painting and music	[K4]
CO5	Analyze Open Source Models and Programming Frameworks for Gen Ai	[K4]

SYLLABUS

UNIT I: Introduction To Gen Ai: Historical Overview of Generative modeling, Difference between Gen AI and Discriminative Modeling, Importance of generative models in AI and Machine Learning, Types of Generative models, GANs, VAEs, autoregressive models and Vector quantized Diffusion models, Understanding if probabilistic modeling and generative process, Challenges of Generative Modeling, Future of Gen AI, Ethical Aspects of AI, Responsible AI, Use Cases.

UNIT II: Generative Models For Text: Language Models Basics, Building blocks of Language models, Transformer Architecture, Encoder and Decoder, Attention mechanisms, Generation of Text, Models like BERT and GPT models, Generation of Text, Autoencoding, Regression Models, Exploring ChatGPT, Prompt Engineering: Designing Prompts, Revising Prompts using Reinforcement Learning from Human Feedback (RLHF), Retrieval Augmented Generation, Multimodal LLM, Issues of LLM like hallucination.

UNIT III: Generation of Images: Introduction to Generative Adversarial Networks, Adversarial Training Process, Nash Equilibrium, Variational Autoencoders, Encoder-Decoder Architectures, Stable Diffusion Models, Introduction to Transformer-based Image Generation, CLIP, Visual Transformers ViT-Dall-E2 and Dall-E3, GPT-4V, Issues of Image Generation models like Mode Collapse and Stability.

UNIT IV: Generation of Painting, Music and Play: Variants of GAN, Types of GAN, Cyclic GAN, Using Cyclic GAN to Generate Paintings, Neural Style Transfer, Style Transfer, Music Generating RNN, MuseGAN, Autonomous agents, Deep Q Algorithm, Actor-critic Network.

UNIT V: Open Source Models and Programming Frameworks: Training and Fine tuning of Generative models, GPT 4 All, Transfer learning and Pretrained models, Training vision models, Google Copilot, Programming LLM, LangChain, Open Source Models, Llama, Programming for TimeSformer, Deployment, Hugging Face.

Text Books:

1. Denis Rothman, “Transformers for Natural Language Processing and Computer Vision”, Third Edition, Packt Books, 2024.

Reference Books:

1. David Foster, “Generative Deep Learning”, O’Reily Books, 2024.
2. Altaf Rehmani, “Generative AI for Everyone”, BlueRose One, 2024.

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Semester	VII	L	T	P	C	COURSE CODE
Regulation	V23	0	0	6	3	V23AITOE08
Name of the Course	**Java Full Stack Technologies** (Open Elective)					
Branch	Except CAI,AIM and CDS					

Course Outcomes: After successful completion of the course, the student will be able to:

CO No.	Course Outcome	Knowledge Level
CO1	Develop programs using Servlets	**[K3]**
CO2	Build applications using MVC architecture	**[K3]**
CO3	Apply Hibernate to simplify database operations	**[K3]**
CO4	Utilize the Spring MVC Framework in web applications	**[K3]**

Syllabus Details

Exercise 1: Understanding about Servlets: Create Example programs Using the below concepts

- Introduction to Servlets.
- Develop Servlet application to print current date & time.
- Develop Servlet program to link Html & Servlet Communication.
- Develop Servlet program to Auto refresh a page.
- Demonstrate session tracking.
- Develop Servlet program to insert/delete/update the record into database.
- Develop Servlet program to add cookie to selected value

Exercise 2: Understanding about Model View Controller: Create Example programs Using the below concepts

- Introduction to MVC in java.
- Develop sample program on Model Layer in MVC Using Java.
- Develop sample program on View Layer in MVC Using Java.
- Develop sample program on Controller Layer in MVC Using Java.
- Demonstrate MVC deployment in java.
- Rules for MVC Mapping in Server Side.
- How to use Web Server for MVC Deployment.

Exercise 3: Understanding about Hibernate: Create Example programs using the below concepts

- Introduction to Hibernate.
- What is ORM? Demonstrate the components of Hibernate
- How to persist objects using Hibernate and how to use map using XML and Annotations
- Working with relationship between entities-association
- Transactions in Hibernate
- Querying with HQL (Hibernate Query Language)
- Various other forms of querying - Criteria, QBE etc.
- Understanding Some Debugging Tools in Java NetBeans, Eclipse, IntelliJ IDEA, Visual Studio Code.

Exercise 4: Develop SpringBoot applications for the following case studies

- **Case Study A: Online Course Management System (OCMS):**Build a scalable Online Course Management System (OCMS) where administrators can upload courses, and students can register, browse, enroll, and track their learning progress.

Core Modules

- **Student Module:** Register and update personal details (name, email, contact),Enroll in courses, Track course progress and view completed courses
- **Course Module:** Create and manage courses (title, description, category, instructor),Assign content like videos, PDFs, quizzes, List all available courses for students

- **Instructor Module:** Register as an instructor, Add and manage courses they're Teaching, View enrolled students and track their performance.
- **Enrollment Module:** Links students to courses, Stores date of enrollment and status (ongoing, completed),Provides history of courses taken by the student.
- **Notification Module:** Sends alerts for new course additions or updates, Reminds students of pending or upcoming coursework, Notifies instructors of newly enrolled students
- **User Authentication & Role Module:** Handles login/signup for both students and instructors, Defines role-based access (e.g. only instructors can create courses), Secures API with Spring Security.

- **Case Study B: Smart Health Monitoring System (SHMS):** Develop a Smart Health Monitoring System for tracking patient vitals (like heart rate, blood pressure, oxygen levels, etc.), enabling doctors to monitor patients remotely in real time.

Core Modules

- **Patient Module:** Register and update personal and medical details
- **Vital Stats Module:** Stores real-time body parameters: `blood Pressure`, `oxygen Level`, `temp`, etc.
- **Alert Module:** Raises alerts if parameters fall outside safe range
- **Doctor Module:** Assigns doctors to patients and shows dashboard

Reference Books:

1. Jeffrey C. Jackson, "Web Technologies-A Computer Science Perspective", Pearson Education, 2006.
2. Murach's Java Servlets and JSP, 3rd Edition by Murach: Training & Reference, 3rd Edition.
3. Spring and Hibernate Paperback – 1 July 2017 by K. Santosh Kumar.
4. Full Stack Java Development with Spring MVC, Hibernate, jQuery, and Bootstrap by Mayur Ramgir, Wiley.

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Semester	VII	L	T	P	C	COURSE CODE
Regulation	V23	0	0	6	3	V23AITOE09
Name of the Course	Agentic AI (Open Elective)					
Branch	Except CAI,AIM and CDS					

Course Outcomes: After successful completion of the course, the student will be able to:

CO No.	Course Outcome	Knowledge Level
CO1	Explain agentic AI concepts, agent loops, tool selection, and query routing	[K2]
CO2	Apply function calling and tool integration to develop single-agent applications.	[K3]
CO3	Develop retrieval and self-correcting agents using LangChain and LangGraph	[K3]
CO4	Construct multi-document Agentic RAG and multi-agent workflows	[K3]
CO5	Develop API-enabled agents with deployment, logging, and basic evaluation	[K3]

Syllabus

Exercise 1: Introduction to Agentic AI + Query Router Agent

Topics: What is an Agent vs Chatbot, Agent loop (observe–decide–act), Tool/function calling concept, Intent classification, Prompts for routing, Action selection logic, Simple agent flow in Python, Campus automation domain framing, Console interaction design, Limitations of basic agents.

Practice: Build a Campus Query Router Agent that classifies a question as Academic, Exam, or General and triggers the respective stub function output.

Assignment: Implement a Python agent that:

- Takes user query
- Uses LLM to classify intent
- Calls the ap
- propriate function (academic/exam/general)
- Prints tool response

Exercise 2: Basic Tool Use & Function Calling for Agents

Topics: System and task prompts, Function-calling concept in agents, Designing and wrapping tools in Python, Reason-to-action execution flow, Classifying user requests into tool categories, Command-line agent interaction, Handling basic math/code/grammar queries, Introducing human-in-loop confirmation logic, Error-safe tool invocation, Understanding limits of rule-based tool routing vs LLM-driven routing

Practice: Build a Student Helper Agent that can understand whether a user needs math help, Python syntax support, or English grammar correction, and then call the appropriate tool function to produce a relevant response.

Assignment: Implement a Python agent that:

- Takes a user query
- Uses an LLM to decide which tool to invoke
- Calls one of the tool functions (math, python_help, english_grammar)
- Prints the selected tool's response to the user

Exercise 3: LangChain Basics & Simple Retrieval Agent

Topics: LangChain introduction, Chains and tools, PromptTemplate basics, Loading text data, Text splitting concepts, Embeddings fundamentals, Vector databases overview (Chroma intro), Document retrieval flow, Retrieval + response pipeline design, Integrating LLM with Python functions, Simple evaluation of retrieved results

Practice: Build a simple Retrieval Agent using LangChain that loads a text document (ex: syllabus or academic guidelines), splits it, embeds it, and answers student questions from that content using a basic retrieval chain.

Assignment: Implement a LangChain-based Python agent that:

- Loads a PDF/text document (campus syllabus or rules)
- Converts it into searchable chunks
- Uses vector search to retrieve the most relevant chunk
- Generates an answer for a student query using the retrieved text

Exercise 4: LangGraph Basics & Self-Correction Loop

Topics: Introduction to LangGraph, State and workflow node concepts, Graph-based agent control, Reflection and retry pattern, Human-approved execution flow, Message/state trimming, Controlled agent pipeline vs direct generation, Designing self-correction cycles, Principles of safe iterative improvement

Practice: Build a Self-Correction Coding Agent using LangGraph that takes a simple coding task, generates code, checks if the output contains an error message, and then revises its response in a second attempt (simulating an autonomous correction loop).

Assignment: Implement a LangGraph agent that:

- Accepts a small programming question (e.g., Python list reverse, factorial, file reading)
- Generates the code answer
- Detects basic errors in its own output (simple text check like "Traceback" or "Error")
- Attempts to correct the response and present the improved code to the user

Exercise 5: Agentic RAG With Structured Retrieval

Topics: RAG vs Agentic RAG, Document chunking strategies, Embedding selection basics, Context-controlled retrieval, Query rewriting behavior, Multi-step retrieval prompts, Preventing irrelevant recall, Citing retrieved sources, Answer confidence estimation, Limitations of naïve RAG and need for agent orchestration

Practice: Build a Campus Information RAG Agent that retrieves content from multiple documents (such as academic handbook, exam rules, and department brochures) and generates context-grounded answers with supporting text snippets.

Assignment: Implement a multi-document RAG system that:

- Loads two or more campus-related documents
- Embeds and stores them in a vector store
- Retrieves context for a student query
- Answers using retrieved text and displays reference snippets

Exercise 6: Multi-Agent Collaboration Basics

Topics: Multi-agent design principles, Agent roles and task delegation, Coordinator vs worker agents, Message-passing fundamentals, Agent role prompting, Task pipeline creation, Handling conflicting outputs, Multi-agent chat flows, Simple evaluation of cooperative agents, Practical multi-agent scenarios in campus settings

Practice: Build a two-agent system where one agent acts as a "Research Agent" to gather bullet points from a topic and a second agent acts as a "Refiner Agent" to turn those points into a well-structured short explanation.

Assignment: Implement a two-agent pipeline that:

- Takes a user topic
- Generates bullet points using the first agent
- Converts them into a short explanatory answer using the second agent
- Delivers the refined output to the user

Exercise 7: API-Enabled Agent for External Automation

Topics: API calls from agents, Python tool execution on demand, Web data extraction basics, JSON response parsing, Email/sheets automation concept, Scheduling simple actions, Safety in executing commands, User approval before execution, Integrating external knowledge sources

Practice: Build a Web-Lookup Agent that fetches simple information from a public API (such as weather, news headline, or university database endpoint) and returns structured results to the user with a short natural-language explanation.

Assignment: Implement an API-powered agent that:

- Receives a query requiring external information
- Calls a public API to retrieve the relevant data
- Parses the response and formats a readable answer
- Displays both raw and explanatory outputs

Exercise 8: Deploying and Evaluating an Agent

Topics: Basic FastAPI integration, Converting agent logic into an API endpoint, HTTP request/response design for agents, Basic logging and trace printing, Response formatting, Monitoring agent behavior, Simple evaluation metrics (accuracy, relevance, latency), Deployment considerations for student projects

Practice: Convert one previous week's agent into a simple web API using FastAPI, invoke it from a browser or command-line client, and observe the structured JSON output along with logs.

Assignment: Deploy an agent as an API that:

- Defines a /query endpoint
- Accepts a user question and routes to agent logic
- Returns the answer in structured JSON format
- Prints logs for each request and response

Textbooks:

1. "An Introduction to MultiAgent Systems", Michael Wooldridge, Wiley, 2nd Ed, 2009.
2. "Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations", Yoav Shoham, Kevin Leyton-Brown, Cambridge University Press, 1st Ed, 2009.
3. "Multiagent Systems", Gerhard Weiss (Ed.), MIT Press, 2nd Ed, 2013.
4. "Designing Data-Intensive Applications", Martin Kleppmann, O'Reilly Media, 1st Ed, 2017.
5. "Generative AI with LangChain: Build production-ready LLM applications and advanced agents usingPython, LangChain, and LangGraph", Ben Auffarth, Leonid Kuligin, Packt Publishing, 2nd Ed, 2025.

References:

1. "LangChain Documentation", https://docs.langchain.com/
2. "LangChain Python Reference", https://reference.langchain.com/python/
3. "LangGraph Documentation", https://langchain-ai.github.io/langgraph/
4. "OpenAI API – Function (Tool) Calling", https://platform.openai.com/docs/guides/function-calling
5. "OpenAI Assistants – Overview", https://platform.openai.com/docs/assistants/overview
6. "CrewAI Documentation", https://docs.crewai.com/
7. "FastAPI – Official Documentation", https://fastapi.tiangolo.com/

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Semester	VII	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE05
Name of the Course	**Prompt Engineering (Skill Enhancement Course)**					
Branch	Common to CSE & CST					

Course Outcomes: After successful completion of the course, the student will be able to:

CO No.	Course Outcome	Knowledge Level
CO1	Apply prompt engineering principles to design and refine effective prompts.	[K3]
CO2	Utilize advanced prompting techniques for different text-based tasks.	[K3]
CO3	Construct structured prompts to generate valid and organized outputs.	[K3]
CO4	Develop a basic RAG application using LangChain and vector stores.	[K3]
CO5	Apply multimodal, agentic, evaluation, and ethical prompting techniques.	[K3]

SYLLABUS

UNIT I: Foundations of Prompt Engineering: Definition of prompt engineering, Distinction between prompt engineering and model fine-tuning, Motivation and benefits of prompt engineering, Core principles of effective prompt design, Anatomy of a prompt, Setting up the Python environment for LLM interaction, Iterative prompting lifecycle, Common prompt pitfalls and remediation

Lab Experiments:

1. Environment & Connectivity: Install required packages (e.g., transformers, openai); securely configure the API key; run a simple "Hello, world" prompt to verify model access.

2. Baseline vs. Enhanced Prompts: Execute a naïve prompt ("Write a one-paragraph bio of Ada Lovelace.") and an enhanced prompt that adds role framing, specificity, and explicit format instructions; compare both outputs for relevance, completeness, and style.

3. Iterative Refinement on a Simple Task: Summarize the plot of the Shakespearean play Romeo and Juliet in two sentences through three rounds of prompt tweaking: a. Minimal instruction. b. Addition of length and style constraints c. Specification of key content elements (setting and theme) Document how each iteration changes and improves the result.

4. Diagnosing Prompt Failures & Edge Cases: Craft a vague or contradictory prompt; analyze the failure mode (ambiguity, missing context, or format errors); refine the prompt by adding examples or clarifying instructions.

UNIT II: Advanced Prompt Patterns & Techniques: Enhanced prompt anatomy: contextual detail and explicit output specifications, Few-shot in-context prompting, Prompt structuring and template design, Role-based prompting to establish personas or system behavior, Negative prompting to filter or suppress undesired content, Constraint specification and instruction enforcement (e.g., length, format), Iterative prompt refinement and optimization

Lab Experiments:

1. Few-Shot vs. Zero-Shot Comparison: Design and execute a zero-shot prompt and a few-shot prompt (with 2–3 exemplar input-output pairs) for a chosen text task (e.g., sentiment classification or translation); compare outputs for accuracy, consistency, and adherence to examples.

2. Role-Based & Negative Prompting: Craft a role-based prompt to establish a specific persona (e.g., "You are a financial advisor…"); then create a negative prompt to suppress undesired content (e.g., "Do not mention any brand names"); evaluate how each influences the model's response.

3. Constraint Specification & Iterative Refinement: Select an open-ended task (e.g., summarizing a technical article); issue a basic prompt; identify failures in length or format; refine the prompt by adding explicit constraints (word count, bullet format, etc.); document improvements over two refinement cycles.

UNIT III: **Structured Output & Reasoning Techniques:** Importance of structured outputs for real-world applications, Prompting for specific formats (lists, tables, Markdown), Generating valid JSON and YAML via explicit instructions, Eliciting chain-of thought reasoning in zero-shot prompts, Decomposing complex tasks into manageable sub tasks

Lab Experiments:

1. Structured Format Prompting: Instruct the model to output information as bullet lists and Markdown tables (e.g., "List three benefits of daily exercise in a Markdown table with columns 'Benefit' and 'Description.'"); verify the output matches the requested structure.

2. JSON/YAML Generation: Provide a brief dataset description (e.g., three books with title, author, publication year) and prompt the model to produce valid JSON or YAML; use a parser to validate syntax and refine the prompt if errors occur.

3. Chain-of-Thought & Task Decomposition: Present a multi-step problem (e.g., a logic puzzle) and apply zero-shot CoT prompting (e.g., "Let's think step by step. Explain your reasoning before the final answer."); separately, decompose the problem into sequential sub-questions, collect partial answers, combine them, and compare accuracy against a direct-answer baseline.

UNIT IV: Retrieval-Augmented Generation & LangChain Workflows: Limitations of LLM internal knowledge, Need for external data sources, Introduction to Retrieval Augmented Generation (RAG), Overview of RAG architecture (indexing vs. retrieval + generation), Getting started with LangChain for LLM applications, Basics of LangChain Expression Language (LCEL), Simplified indexing pipeline: document loading & text splitting, Fundamentals of embeddings and vector stores, Building a basic retrieval generation pipeline with an LCEL chain

Lab Experiments:

1. Building a Simple LCEL Chain: Create a minimal LCEL script that accepts a fixed instruction (e.g., "Summarize this text: …"), passes it to an LLM, and prints the result; verify end-to-end execution.

2. Basic Data Indexing for RAG: Load a small collection of documents; split into uniform chunks (e.g., 200 tokens); generate embeddings for each chunk; store them in an in-memory vector store; inspect for consistency.

3. Constructing & Running a Basic RAG Chain: Build a pipeline that:
   a. Receives a user query
   b. Retrieves the top-k relevant chunks
   c. Constructs a combined prompt with context + query
   d. Send it to the LLM
   e. Returns the answer

Test with sample queries and compare factual accuracy against a prompt without retrieval.

UNIT V: Agents, Multimodal AI & Ethical Evaluation: Introduction to LLM agents and their basic architecture, Overview of multimodal AI models (VLMs), Prompting for text-to image generation and image understanding, Importance of prompt evaluation beyond subjective judgment, Manual evaluation techniques (heuristic checks for accuracy, relevance, format), Introduction to "LLM-as-Judge" for automated evaluation, Security considerations (prompt injection, sensitive-information risks), Prompt-

based mitigation strategies for safety and robustness, Ethical concerns (bias, misinformation, data privacy), Brief exploration of UI frameworks (Streamlit/Gradio) for deploying prompt-driven apps, Adapting to the evolving nature of prompt engineering through continuous learning

Lab Experiments:

1. Building a Simple LLM Agent: Register a tool (e.g., a calculator function) and craft prompts that instruct the agent to invoke it when required; implement using LangChain or a function-calling API; test on queries requiring tool execution.

2. Multimodal Prompting Exploration: Generate images from detailed text prompts; feed one generated image into an image-understanding model or API with an appropriate prompt; compare the returned caption to the original prompt to evaluate alignment.

3. Prompt Evaluation & Ethics Workshop:
   a. Select two existing prompts and generate multiple outputs; apply manual heuristic checks for accuracy, relevance, and format compliance.
   b. Use an "LLM-as-Judge" prompt (e.g., "Rate these outputs on a scale of 1–5 for clarity and correctness.") to automate evaluation.
   c. Design a prompt- injection test (e.g., "Ignore previous instructions…"), observe the response, then refine system prompts to mitigate the vulnerability.

Reference Books:

1. The Art of Prompt Engineering with ChatGPT, Nathan Hunter, 1st Edition, Sharoff.
2. Designing Intelligent Systems Using AI and LLMs, Ben Deloose, 1st Edition, Packt.

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Semester	VII	L	T	P	C	COURSE CODE
Regulation	V23	2	0	0	0	
Name of the Course	AI Ethics and Cyber Laws **(Audit Course)**					
Branch	Common to CSE & CST					

Course Outcomes: After successful completion of the course, the student will be able to:

CO No.	Course Outcome	Knowledge Level
CO1	Explain basic AI concepts and ethics	[K2]
CO2	Illustrate ethical issues and risks of AI.	[K2]
CO3	Explain basic cyber laws and digital rights.	[K2]
CO4	Explain cybercrimes and safe internet practices.	[K2]
CO5	Outline AI governance, quantum computing, and future digital challenges.	[K2]

SYLLABUS

UNIT I: Introduction to AI and Ethics:
- What is AI?
- Everyday applications, Need for ethics in AI.
- Basic concepts: fairness, bias, transparency.

UNIT II: AI in Society
- AI in social media, healthcare, and education
- Privacy concerns and data misuse
- Real-life examples of AI risks

UNIT III: Basics of Cyber Laws
- Introduction to cyber law
- Overview of Information Technology Act, 2000
- Digital identity and online rights

UNIT IV: Cyber Safety and Cybercrime
- Common cybercrimes: phishing, hacking, fraud
- Safe internet practices
- Reporting cybercrime and basic legal awareness

UNIT V: Emerging Trends (Overview)
- AI governance (basic idea)
- Introduction to quantum computing (awareness level only)
- Future digital challenges

Textbooks:

1. An Introduction to Cyber Law, Vivek Sood, Latest Edition, 2017, (**Unit 3, Unit 4)**
2. AI Ethics, Mark Coeckelbergh, 1st Edition, 2020, (**Unit 1, Unit 2, Unit 5)**

Reference Books:

1. Cyber Law, Pavan Duggal, 3rd Edition, 2025, (Unit 3, Unit 4)
2. The Ethics of Artificial Intelligence, S. Matthew Liao, 1st Edition, 2020, **(Unit 1, Unit 2, Unit 5)**
3. Quantum Computing for Everyone, Chris Bernhardt, 1st Edition, 2019,(**Unit 5)**

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